[21/11/2025]

1. Scope of Work

Program Overview

This project aims to enhance cybersecurity threat intelligence workflows through student capstone projects. Specifically, the project will involve the development of AI tools as a part of student capstone projects that can summarize the threat landscape reports, extract key information, and generate concise and readable narratives. This will assist investigators in quickly determining whether a newly observed threat has been previously documented in event logs or historical reports, thereby improving investigative efficiency and decision-making capabilities.

The Envisioned Usage and Functions of an AI tool would be:

Usage

- Enrich raw cybersecurity data, including event logs, vulnerabilities, and threat intelligence reports.
- Facilitate faster threat detection, correlation, and historical reference during security investigations.
- Intended solely for internal use in cybersecurity operations, supporting the enhancement of investigative workflows for security teams.

Functions

- Automated summarization of Threat landscape reports utilizing the pre-trained summarization models or AI APIs to convert long threat landscape reports into concise actionable narratives.
- Extracting key points from source materials and generating an easy-to-read narrative through off-the-shelf and/or custom NLP tools.
- Enabling searchability across historical event logs and threat reports to identify previously observed threats through the integration of AI tools and APIs.
- Providing enriched contextual analysis for faster incident response and threat attribution through correlation of new incidents with historical events and known TTPs from internal threat databases or local MITRE ATT&CK datasets.

Relevant Details

Datasets:

The project will utilize publicly available, open-source cybersecurity datasets for training and enrichment purposes. These include:

- Vulnerability and threat intelligence databases such as the Common Vulnerabilities and Exposures (CVE), National Vulnerability Database (NVD), and datasets from Shadow server Foundation.
- Publicly available cybersecurity reports published by the European Union Agency for Cybersecurity (ENISA), the National Institute of Standards and Technology (NIST), Microsoft, AT&T, and other recognized organizations.

Language:

The language of this project is spoken/written English.

Agreement does not create binding rights or obligations until signed and dated by an authorized representative from each Party.